MICROSOFT CERTIFICATION PREP

AI-901: Microsoft Azure AI Fundamentals

The Complete Visual Study Guide

120 Slides · Every Domain, Every Concept, Every Mnemonic You Need to Pass

WELCOME

How to Use This Guide

1. **Read actively.** Each concept slide pairs a definition with an exam-relevant example — don't just skim, picture the scenario.

2. **Trust the mnemonics.** Gold “Memory Booster” slides appear at regular intervals. They compress whole lists into one recallable phrase.

3. **Practice as you go.** Practice Check slides simulate real exam wording. Cover the answer and reason it out first.

4. **Follow the two domains.** Everything maps to Microsoft's official blueprint: AI Concepts & Capabilities, then Implementing Solutions with Microsoft Foundry.

5. **Review, don't reread.** In your final pass, jump straight to the gold mnemonic slides and the practice checks — they are your fastest recall path.

EXAM AT A GLANCE

Know the Test Before You Take It

40–60

Questions

Mix of multiple-choice and multiple-response items

45–65

Minutes

Timed exam window at a Pearson VUE testing center

700

Passing Score

Scaled score out of 1000; no partial credit per topic

2

Skill Domains

AI Concepts (40–45%) and Microsoft Foundry (55–60%)

AI-901 replaced AI-900 in 2026 — same beginner-friendly spirit, with far more hands-on Microsoft Foundry content.

YOUR ROADMAP

A Simple 4-Step Study Plan

1

Learn the Concepts

Work through Domain 1: AI, ML, Responsible AI, and every AI workload type.

2

Learn the Platform

Work through Domain 2: Microsoft Foundry, prompting, agents, and the SDK.

3

Drill the Mnemonics

Revisit every gold memory-booster slide until each phrase is automatic.

4

Simulate the Exam

Answer every Practice Check slide cold, then review the explanations.

Most successful candidates spend 1–2 weeks moving through these four stages before sitting the exam.

THE BLUEPRINT

What Microsoft Actually Tests

1. **Domain 1 — Identify AI Concepts & Capabilities (40–45%):** Responsible AI principles, how generative AI and models work, and the five core AI workloads (text, speech, vision, image generation, information extraction).

2. **Domain 2 — Implement AI Solutions Using Microsoft Foundry (55–60%):** The Foundry portal, model selection and deployment, prompt engineering, the Foundry SDK, building agents, and responsible-AI tooling in practice.

3. **Why it's weighted this way:** AI-901 is a hands-on evolution of AI-900 — Microsoft wants candidates who can actually build a lightweight AI solution, not just define terms.

Keep this ratio in mind: for every 2 questions on pure concepts, expect roughly 3 on applying Foundry.

PART ONE

AI Concepts & Capabilities

Foundations of artificial intelligence, machine learning, responsible AI, and the five core AI workloads tested on the exam.

Domain Weight: 40–45% of the exam

FOUNDATIONS

What Is Artificial Intelligence?

1. **Definition:** Software that performs tasks normally requiring human intelligence — perceiving, reasoning, learning, and deciding.

2. **It's a spectrum, not a single tool:** From simple rule-based automation to systems that learn patterns from data and adapt over time.

3. **The exam framing:** Microsoft cares less about sci-fi AI and more about practical AI workloads you can build on Azure and Microsoft Foundry.

4. **Key mental model:** AI is the broad goal; Machine Learning is the dominant technique used to achieve it today.

THE NESTED DOLLS

AI, Machine Learning & Deep Learning

1

Artificial Intelligence

- The broadest field
- Any technique that mimics intelligent behavior
- Includes rule-based systems too

2

Machine Learning

- A subset of AI
- Learns patterns from data
- Improves with more examples, not more rules

3

Deep Learning

- A subset of ML
- Uses multi-layer neural networks
- Powers modern generative AI

Three Types of Machine Learning

1

Supervised Learning

- Trained on labeled data
- Learns input → output mapping
- Used for regression & classification

2

Unsupervised Learning

- Trained on unlabeled data
- Finds hidden structure
- Used for clustering & anomaly detection

3

Reinforcement Learning

- Learns via trial and reward
- An agent acts in an environment
- Used for optimization & control

SUPERVISED LEARNING

Learning From Labeled Examples

1. **Regression:** Predicts a continuous numeric value — e.g., forecasting a house price or tomorrow's temperature.

2. **Classification:** Predicts a category or class — e.g., “spam vs. not spam,” or “will this loan default?”

3. **How it trains:** The model sees many (input, correct-answer) pairs and adjusts itself to minimize prediction error.

4. **Exam cue:** If the scenario mentions historical labeled outcomes used to predict a future outcome, think supervised learning.

UNSUPERVISED LEARNING

Finding Structure Without Labels

1. **Clustering:** Groups similar data points together — e.g., segmenting customers by purchasing behavior.

2. **Dimensionality reduction:** Simplifies data while preserving its important structure, often used before visualization.

3. **Anomaly detection:** Flags data points that don't fit the learned pattern — e.g., fraudulent transactions.

4. **Exam cue:** If the scenario has no “correct answer” to learn from, and just wants patterns or groups, think unsupervised learning.

REINFORCEMENT LEARNING

Learning by Reward and Consequence

1. **The loop:** An agent takes an action, the environment responds with a new state and a reward or penalty, and the agent adjusts its strategy.

2. **The goal:** Maximize cumulative reward over time — not just get one answer right.

3. **Classic use cases:** Game-playing AI, robotics, resource optimization, and tuning recommendation strategies.

4. **Exam cue:** Look for language about an “agent,” “environment,” and “reward” — that combination signals reinforcement learning.

MEMORY BOOSTER

Recall the 3 Machine Learning Types

“Sam Uses Rewards”

S	**Supervised**	Sam (a teacher) gives labeled examples — the model learns from correct answers.
U	**Unsupervised**	Uses no labels — the model discovers its own groupings and patterns.
R	**Reinforcement**	Rewards guide the agent — it learns through trial, error, and feedback.

Say it once out loud — “Sam Uses Rewards” — and you'll never mix up the three ML types again.

GOING DEEPER

Neural Networks & Deep Learning

1. **Neural network basics:** Layers of connected “neurons” that transform input data through weighted connections into an output.

2. **Why “deep” matters:** Many stacked hidden layers let the network learn increasingly abstract features — edges, then shapes, then objects.

3. **Training:** Weights are adjusted through backpropagation to reduce the difference between predicted and actual outputs.

4. **Why it matters for AI-901:** Deep learning architectures (especially transformers) are the engine behind every generative AI model on Foundry.

THE MODERN WAVE

What Is Generative AI?

1. **Definition:** AI that creates new content — text, images, audio, or code — rather than just classifying or predicting existing data.

2. **The shift from traditional ML:** Traditional ML predicts a label or number; generative AI produces original, human-like output.

3. **Powered by:** Large-scale deep learning models trained on massive datasets, most commonly built on the transformer architecture.

4. **Exam cue:** Any scenario about drafting text, generating images, or holding a natural conversation points to generative AI.

THE MODELS BEHIND IT

Foundation Models & Large Language Models

1. **Foundation model:** A large, general-purpose model pre-trained on broad data, adaptable to many downstream tasks without starting from scratch.

2. **Large Language Model (LLM):** A foundation model specialized in understanding and generating human language.

3. **Small Language Model (SLM):** A lighter-weight model, cheaper and faster to run, trading some capability for efficiency — great for edge or cost-sensitive scenarios.

4. **Why this matters for Foundry:** Choosing between an LLM and an SLM is a real exam scenario — you must match model size to the task's complexity and budget.

MODEL LIFECYCLE

Foundation Model vs. Fine-Tuned Model

Foundation Model (Base)

- Pre-trained on massive, general datasets
- Broad, general-purpose capability
- Used directly via prompting
- Fast to start, no extra training needed
- Best for general tasks and prototyping

Fine-Tuned Model

- Further trained on a smaller, specific dataset
- Specialized for a narrow domain or style
- Adjusts model weights to your data
- Requires curated training examples
- Best for consistent tone, format, or niche tasks

VOCABULARY CHECK

Core LLM Building Blocks

1 Token

A chunk of text (word or word-piece) the model reads and generates — the basic unit of language processing.

2 Embedding

A numeric vector representing the meaning of a token so machines can compare and process it mathematically.

3 Parameters

The internal weights learned during training — more parameters generally means more capacity to capture patterns.

4 Context Window

The maximum amount of text (in tokens) a model can “see” at once, including prompt and conversation history.

UNDER THE HOOD

How LLMs Actually Generate Text

1. **Next-token prediction:** The model repeatedly predicts the single most likely next token given everything written so far.

2. **The transformer architecture:** Uses an “attention” mechanism to weigh how relevant every earlier token is to predicting the next one.

3. **It's probabilistic, not magic:** Outputs are statistically likely continuations — which is exactly why generative AI can be fluent yet factually wrong.

4. **Exam cue:** “Hallucination” refers to confident but incorrect output — a direct consequence of this prediction-based process.

PRACTICE CHECK 1

A retailer wants a system that studies five years of labeled sales records to predict next month's revenue. Which approach fits best?

A Unsupervised learning — clustering similar sales records together

B Supervised learning (regression) — trained on historical labeled outcomes ✓ CORRECT

C Reinforcement learning — an agent rewarded for accurate guesses

D Generative AI — drafting a written sales forecast narrative

Why: Predicting a continuous numeric value (revenue) from labeled historical data is a textbook regression problem — a form of supervised learning.

RESPONSIBLE AI

Why Responsible AI Matters

1. **AI can scale mistakes fast:** A biased or unsafe model doesn't affect one decision — it affects every decision it's used for, at machine speed.

2. **Microsoft's commitment:** Every Azure AI and Foundry service is built around a formal set of Responsible AI principles, not treated as an afterthought.

3. **The exam weight:** Responsible AI questions appear throughout Domain 1 — expect scenario-based questions, not just definitions.

4. **The six principles preview:** Fairness, Reliability & Safety, Privacy & Security, Inclusiveness, Transparency, and Accountability.

PRINCIPLE 1 OF 6

Fairness

1. **What it means:** AI systems should treat all people fairly and not create or reinforce unfair bias based on gender, ethnicity, or other characteristics.

2. **Where bias creeps in:** Training data that under-represents a group, or historical data that encodes past human bias.

3. **Azure tooling:** Fairness assessment components in the Responsible AI dashboard measure disparity in model outcomes across groups.

4. **Exam cue:** A scenario about a hiring or lending model producing unequal outcomes across demographic groups tests Fairness.

PRINCIPLE 2 OF 6

Reliability & Safety

1. **What it means:** AI systems should perform consistently, as intended, and handle unexpected conditions gracefully — especially in high-stakes settings.

2. **High-stakes examples:** Autonomous vehicles, medical diagnosis assistance, and safety-critical industrial systems demand rigorous testing.

3. **Best practice:** Extensive testing and a human-in-the-loop safety net before and after deployment.

4. **Exam cue:** Any scenario about testing rigor, edge-case handling, or system failure risk points to Reliability & Safety.

PRINCIPLE 3 OF 6

Privacy & Security

1. **What it means:** AI solutions must protect personal data and defend against vulnerabilities across its lifecycle.

2. **Privacy focus:** Collect and use only the data that's necessary; give people control and transparency over their own data.

3. **Security focus:** Guard the model, data pipeline, and endpoints against unauthorized access, tampering, or misuse.

4. **Exam cue:** Data anonymization, consent, or access-control language signals Privacy & Security.

PRINCIPLE 4 OF 6

Inclusiveness

1. **What it means:** AI should empower and engage everyone, including people with disabilities and diverse backgrounds, languages, and experiences.

2. **In practice:** Designing for accessibility — alt text, speech interfaces, multilingual support — from the start, not bolted on later.

3. **Business case:** Inclusive design widens who can actually benefit from and adopt the solution.

4. **Exam cue:** Accessibility features or serving underrepresented user groups points to Inclusiveness.

PRINCIPLE 5 OF 6

Transparency

1. **What it means:** People should understand how and why an AI system makes its decisions, and know when they are interacting with AI.

2. **In practice:** Disclosing AI involvement, documenting model capabilities and limitations, and explaining key factors behind a decision.

3. **Azure tooling:** Interpretability components in the Responsible AI dashboard help explain individual model predictions.

4. **Exam cue:** “Explainability,” “disclosure,” or “users should know they're talking to AI” all map to Transparency.

PRINCIPLE 6 OF 6

Accountability

1. **What it means:** People — not the algorithm — remain responsible for how an AI system behaves and the outcomes it produces.

2. **In practice:** Clear governance structures, human oversight, and the ability to intervene or reverse an automated decision.

3. **The connecting thread:** Accountability is what ties the other five principles together — someone must own the outcome.

4. **Exam cue:** Governance, oversight boards, or “who is responsible when it goes wrong” questions map to Accountability.

MEMORY BOOSTER

Recall All 6 Responsible AI Principles

“Fair Robots Protect Innocent Trusting Allies”

F	**Fairness**	Treat every person and group equitably — no unfair bias.
R	**Reliability & Safety**	Perform consistently and safely, even in edge cases.
P	**Privacy & Security**	Protect personal data and defend against misuse.
I	**Inclusiveness**	Empower everyone, including people with disabilities.
T	**Transparency**	Make AI decisions understandable and disclosed.
A	**Accountability**	People stay responsible for AI outcomes.

F-R-P-I-T-A — six letters, six principles, one sentence you'll never forget on exam day.

PUTTING IT INTO PRACTICE

Responsible AI: Do's and Don'ts

Do

- Test models across diverse demographic groups
- Disclose when users are interacting with AI
- Document model limitations and intended use
- Keep a human in the loop for high-stakes decisions
- Limit data collection to what's truly necessary

Don't

- Deploy a model without bias or fairness testing
- Let AI make irreversible decisions with no oversight
- Hide AI involvement from end users
- Ignore edge cases because “it works most of the time”
- Collect more personal data than the task requires

PRACTICE CHECK 2

A company deploys a resume-screening model and later discovers it consistently ranks certain demographic groups lower. Which Responsible AI principle was violated?

A Transparency — the model's logic wasn't explained

B Fairness — the model produced unequal outcomes across groups ✓ CORRECT

C Inclusiveness — the interface wasn't accessible

D Privacy & Security — candidate data was exposed

Why: Unequal treatment or outcomes across demographic groups is the defining signal of a Fairness violation, regardless of intent.

TOOLING

Governance Tools That Support Responsible AI

1 Responsible AI Dashboard

A unified view combining fairness, error analysis, interpretability, and causal analysis for a deployed model.

2 Content Filters

Configurable safety layers in Foundry that block harmful, violent, or otherwise disallowed generated content.

3 Model Cards

Documentation describing a model's intended use, limitations, and evaluation results — supports Transparency.

4 Human-in-the-Loop Review

A workflow step where a person reviews or approves AI output before it takes effect — supports Accountability.

THE FIVE PILLARS

The Core AI Workloads Tested on AI-901

1. **1. Text Analysis (NLP):** Understanding, extracting meaning from, and processing written language.
2. **2. Speech:** Converting spoken language to text and text back into natural speech.
3. **3. Computer Vision:** Understanding and interpreting the content of images and video.
4. **4. Image Generation:** Creating new, original images from text prompts or reference input.
5. **5. Information Extraction:** Pulling structured data — fields, tables, key-value pairs — out of unstructured documents.

Each of the next sections goes one workload deep — know the core task names, not just the category.

TEXT ANALYSIS

What Is Natural Language Processing?

1. **Definition:** The branch of AI focused on enabling computers to read, interpret, and generate human language.

2. **Two directions:** Natural Language Understanding (NLU) extracts meaning; Natural Language Generation (NLG) produces new text.

3. **Where it shows up:** Chatbots, search engines, content moderation, customer feedback analysis, and document processing.

4. **Exam framing:** AI-901 tests whether you can name the right NLP task for a given business scenario.

TEXT ANALYSIS

Key Phrase & Entity Extraction

1. **Key phrase extraction:** Identifies the main talking points in a block of text — useful for quickly summarizing themes.

2. **Named entity recognition (NER):** Detects and categorizes real-world objects mentioned in text — people, places, organizations, dates, quantities.

3. **Entity linking:** Connects a recognized entity to a knowledge base entry to disambiguate it — e.g., distinguishing "Mercury" the planet from the element.

4. **Exam cue:** "Pull out names, dates, and locations" is entity recognition; "summarize the main topics" is key phrase extraction.

TEXT ANALYSIS

Sentiment Analysis & Opinion Mining

1. **Sentiment analysis:** Classifies text as positive, negative, neutral, or mixed — often with a confidence score.

2. **Opinion mining:** A finer-grained version that links sentiment to specific aspects — e.g., “battery life: negative, screen: positive.”

3. **Common use case:** Analyzing product reviews or support tickets at scale to spot trends humans would miss.

4. **Exam cue:** Any scenario measuring how customers “feel” about something maps to sentiment analysis.

TEXT ANALYSIS

Language Detection & Translation

1. **Language detection:** Identifies which language a piece of text is written in, often with a confidence score.

2. **Machine translation:** Converts text from a source language into a target language while preserving meaning.

3. **Why detection comes first:** Most translation and analysis pipelines detect the language before choosing the right downstream model.

4. **Exam cue:** A multinational support desk auto-routing tickets by language is a language-detection scenario.

TEXT ANALYSIS

Summarization

1. **Purpose:** Condenses a long document into a shorter version while preserving the essential meaning.

2. **Two techniques exist:** Extractive summarization and abstractive summarization — covered side-by-side next.

3. **Common scenario:** Summarizing lengthy call-center transcripts or legal documents into a short brief for a human reviewer.

4. **Exam cue:** Look for whether the exam wants a “short version using the original wording” (extractive) or a “rewritten short version” (abstractive).

TWO SUMMARIZATION STYLES

Extractive vs. Abstractive Summarization

Extractive

- Selects existing sentences verbatim
- Stitches the most important ones together
- Guaranteed to use original wording
- Generally faster and less “creative”
- Lower risk of introducing new errors

Abstractive

- Generates brand-new sentences
- Paraphrases and synthesizes meaning
- Reads more naturally, like a human summary
- Requires a generative language model
- Higher risk of subtle inaccuracies

MEMORY BOOSTER

Recall the Core Text Analysis Tasks

“Keep Every Sentence Loud & True”

K	**Key Phrase Extraction**	Surfaces the main talking points in a passage.
E	**Entity Recognition**	Finds people, places, organizations, and dates.
S	**Sentiment Analysis**	Scores text as positive, negative, or neutral.
L	**Language Detection**	Identifies which language the text is written in.
T	**Translation & Summarization**	Converts languages, or condenses long text.

“Keep Every Sentence Loud & True” covers the five text-analysis tasks the exam loves to test.

PRACTICE CHECK 3

A support team wants a dashboard that automatically flags angry customer emails for priority review. Which NLP capability fits best?

A Named entity recognition

B Language detection

C Sentiment analysis ✓ CORRECT

D Extractive summarization

Why: Detecting “anger” in the tone of a message is a sentiment classification task — exactly what sentiment analysis is built for.

Speech Recognition (Speech-to-Text)

1. **Definition:** Converts spoken audio into written text in real time or from a recorded file.

2. **How it works:** An acoustic model maps sound waves to phonemes, while a language model predicts the most likely word sequence.

3. **Real-world uses:** Live captioning, meeting transcription, voice commands, and dictation.

4. **Exam cue:** “Turn a customer call into a text transcript” is a speech-to-text scenario.

SPEECH

Speech Synthesis (Text-to-Speech)

1. **Definition:** Converts written text into natural-sounding spoken audio.

2. **Neural voices:** Modern text-to-speech uses neural networks to produce human-like intonation, pacing, and emotion.

3. **Customization:** Custom neural voice lets an organization build a unique, brand-specific voice from voice samples.

4. **Exam cue:** “Read this article aloud” or “give our IVR system a natural voice” points to text-to-speech.

TWO DIRECTIONS OF SPEECH

Speech-to-Text vs. Text-to-Speech

Speech-to-Text

- Input: audio → Output: text
- Use case: live meeting captions
- Use case: voice-controlled assistants
- Use case: call center transcription
- Key metric: word error rate

Text-to-Speech

- Input: text → Output: audio
- Use case: audiobook narration
- Use case: accessibility screen readers
- Use case: IVR & voice assistants
- Key feature: neural, natural-sounding voices

BEYOND THE BASICS

Additional Speech Capabilities

1 Speech Translation

Translates spoken audio in one language directly into text or speech in another language.

2 Speaker Recognition

Identifies or verifies who is speaking, based on unique vocal characteristics.

3 Custom Voice

Trains a distinctive, brand-specific synthetic voice from recorded voice samples.

4 Pronunciation Assessment

Scores how accurately a speaker pronounces words — useful for language-learning apps.

PRACTICE CHECK 4

A language-learning app needs to score how accurately a student pronounces new vocabulary words aloud. Which speech capability applies?

A. Speaker recognition

B. Pronunciation assessment ✓ CORRECT

C. Speech translation

D. Custom neural voice

Why: Pronunciation assessment is purpose-built to score accuracy, fluency, and completeness of spoken pronunciation against a reference.

What Is Computer Vision?

1. **Definition:** The field of AI that enables computers to interpret and understand visual information from images and video.

2. **Core idea:** Vision models turn pixels into structured, usable information — labels, coordinates, or extracted text.

3. **Where it shows up:** Retail shelf monitoring, manufacturing defect detection, medical imaging, and accessibility tools.

4. **Exam framing:** You need to match a vision scenario to the correct specific task — covered one by one next.

COMPUTER VISION

Image Classification

1. **Definition:** Assigns one or more labels to an entire image, describing what it depicts overall.

2. **Single-label vs. multi-label:** An image can be classified into one category (“cat”) or several simultaneously (“beach, sunset, ocean”).

3. **Custom vision models:** You can train a classifier on your own labeled images for a domain-specific task.

4. **Exam cue:** “Is this a picture of a defective part or a good part?” is classic image classification.

Object Detection

1. **Definition:** Identifies multiple objects within an image and draws a bounding box around each one.

2. **Classification vs. detection:** Classification answers “what is in this image?”; detection answers “what is in this image, and where?”

3. **Common uses:** Counting inventory on a shelf, detecting vehicles in traffic footage, or locating safety-gear violations.

4. **Exam cue:** Any scenario needing the position or count of multiple items points to object detection.

Optical Character Recognition (OCR)

1. **Definition:** Extracts printed or handwritten text from images and scanned documents into machine-readable text.

2. **Where it fits:** The bridge between computer vision and text analysis — OCR output often feeds directly into NLP tasks.

3. **Common uses:** Digitizing receipts, reading license plates, and converting scanned forms into searchable text.

4. **Exam cue:** “Read the text in this photo or scanned document” always maps to OCR.

COMPUTER VISION

Facial Detection & Responsible Use

1. **Facial detection:** Locates human faces within an image, typically returning a bounding box for each face.

2. **Facial analysis:** Can estimate attributes such as approximate age range or emotion — always with accuracy caveats.

3. **Facial recognition (identity matching) is restricted:** Microsoft applies strict Responsible AI gating — limited access, required use-case review, and no unrestricted public deployment.

4. **Exam cue:** Expect a question testing that facial recognition is treated as the most sensitive vision capability under Responsible AI.

MORE VISION CAPABILITIES

Spatial Analysis & Other Vision Tasks

1 Spatial Analysis

Understands people's movement and position in physical space from video — e.g., counting people in a zone, or social distancing.

2 Image Segmentation

Classifies every individual pixel, producing a precise outline of each object rather than just a bounding box.

3 Image Analysis (Captioning)

Generates a natural-language description of what an image depicts.

4 Video Indexing

Extracts insights across a video timeline — detected objects, spoken transcript, and scene changes.

MEMORY BOOSTER

Recall the Core Computer Vision Tasks

“Clever Owls Detect Objects”

C	**Classification**	Labels the whole image with one or more categories.
O	**OCR**	Reads printed or handwritten text out of an image.
D	**Detection**	Finds multiple objects and boxes their exact location.
O	**(Facial) Overview**	Detects and analyzes faces — handled with extra Responsible AI care.

“Clever Owls Detect Objects” anchors the four vision tasks the exam tests most often.

PRACTICE CHECK 5

A warehouse camera feed needs to count how many forklifts are visible and mark each one's exact location in the frame. What's the right capability?

A. Image classification

B. Object detection ✓ CORRECT

C. Optical character recognition

D. Sentiment analysis

Why: Counting multiple items and locating each one within the frame is exactly what object detection's bounding boxes provide.

Generative Image Models

1. **Definition:** Models that create brand-new images from a text description (text-to-image) or from a reference image.

2. **How it differs from CV analysis:** Computer vision interprets existing images; image generation produces original ones.

3. **Common capabilities:** Text-to-image generation, image editing/inpainting, and style transfer.

4. **Exam cue:** “Create marketing artwork from a written brief” is an image-generation scenario, not computer vision.

Responsible Use of Image Generation

1. **Deepfake and misinformation risk:** Generated images can be used to fabricate misleading or harmful visual content.

2. **Content provenance:** Techniques like watermarking and metadata (e.g., Content Credentials) help mark AI-generated images as such.

3. **Content filtering:** Foundry-based image models apply moderation to block disallowed prompts and outputs.

4. **Exam cue:** Image generation scenarios often test the Fairness and Transparency principles from Responsible AI.

TWO SIDES OF VISION AI

Image Generation vs. Image Analysis

Image Generation

- Input: text prompt → Output: new image
- Creates content that didn't exist before
- Use case: concept art, marketing visuals
- Key risk: misuse, deepfakes

Image Analysis (Computer Vision)

- Input: existing image → Output: labels/data
- Interprets content that already exists
- Use case: classification, detection, OCR
- Key risk: bias in labeling or recognition

INFORMATION EXTRACTION

Pulling Structure Out of Documents

1. **Definition:** AI capability that identifies and extracts specific structured data points from unstructured documents.

2. **Why it's its own workload:** It combines OCR (reading the text) with layout understanding (knowing where a field sits on the page).

3. **Common uses:** Extracting totals from invoices, fields from ID cards, or clauses from contracts.

4. **Exam cue:** “Automatically pull the vendor name and total from thousands of invoices” is an information-extraction scenario.

Document Intelligence & Content Understanding

1. **Document Intelligence:** Azure's prebuilt and custom models for extracting text, key-value pairs, tables, and structure from documents.

2. **Content Understanding (in Foundry):** A newer Foundry tool that extends extraction to multimodal content — documents, images, audio, and video — into a structured output.

3. **Prebuilt vs. custom models:** Prebuilt models handle common document types (invoices, receipts, IDs) out of the box; custom models are trained on your specific form layout.

4. **Exam cue:** If the document type is common (invoice, receipt, ID), reach for a prebuilt model before building a custom one.

WHAT GETS EXTRACTED

Types of Extracted Data

1 Key-Value Pairs

Labeled fields like “Invoice Number: 4471” extracted as a matched pair.

2 Tables

Structured rows and columns preserved from the original document layout.

3 Layout & Structure

Paragraphs, headings, and reading order preserved for downstream processing.

4 Prebuilt Models

Ready-to-use models for invoices, receipts, IDs, business cards, and tax forms.

AGENTIC AI

What Is Agentic AI?

1. **Definition:** AI systems that can plan, decide, and take multi-step action toward a goal — not just respond to a single prompt.

2. **The key shift:** A chat model answers a question; an agent can call tools, retrieve information, and take actions to complete a task.

3. **Why it matters now:** Agentic AI is the headline addition in AI-901 versus the older AI-900 exam — expect it tested heavily in Domain 2.

4. **Exam framing here:** Domain 1 tests the concept; Domain 2 tests actually building an agent in Microsoft Foundry.

AI Agents vs. Simple Chatbots

Simple Chatbot

- Responds to a single prompt at a time
- No memory of taking real-world action
- Limited to generating text replies
- Good for FAQ and simple Q&A

AI Agent

- Plans and executes multi-step tasks
- Can call tools, functions, and APIs
- Retrieves knowledge to ground its answers
- Good for booking, research, and workflows

AI Model Components

1. **Model weights:** The learned numeric parameters that encode everything the model picked up during training.

2. **Inference:** The process of running a trained model on new input to produce an output — this is what happens every time you send a prompt.

3. **Deployment:** Publishing a trained model behind an endpoint so applications can call it over the network.

4. **Why this matters:** AI-901 expects you to know that training happens once (mostly), but inference happens on every single request.

MODEL CONFIGURATION

Configuring a Model's Behavior

1. **Why configure at all:** The same underlying model can behave very differently depending on the parameters you set at inference time.

2. **No retraining required:** These settings are applied per-request — you're not changing the model's weights, just how it samples output.

3. **Where you set them:** In the Foundry playground, SDK code, or API request body.

4. **Coming up next:** The four parameters tested most often — temperature, top-p, max tokens, and the system message.

THE FOUR KEY DIALS

Common Model Parameters

1 Temperature

Controls randomness. Low = focused and predictable; high = creative and varied output.

2 Top P (nucleus sampling)

Limits word choices to the smallest set whose combined probability exceeds P — another creativity dial.

3 Max Tokens

Caps the length of the generated response, controlling cost and preventing runaway output.

4 System Message

Sets the model's persona, tone, and behavioral rules before the conversation begins.

MEMORY BOOSTER

Recall the Key Model Parameters

“Tame The Model's Speech”

T	**Temperature**	Tames or unleashes randomness — low is safe, high is creative.
T	**Top P**	Trims the pool of likely next words to the most probable set.
M	**Max Tokens**	Marks the ceiling on how long a response can run.
S	**System Message**	Sets the model's role, tone, and ground rules upfront.

“Tame The Model's Speech” — Temperature, Top P, Max tokens, System message.

Model Deployment Options

Serverless (Pay-as-you-go) API

- No infrastructure to manage
- Billed per token/request consumed
- Fastest way to get started
- Best for variable or low-to-medium traffic

Managed Compute (Dedicated)

- Reserved GPU/CPU capacity you provision
- Predictable billing, predictable latency
- More control over scaling and isolation
- Best for high, steady-volume workloads

PRACTICE CHECK 6

A developer wants an assistant that gives very consistent, low-variation answers for a compliance FAQ tool. Which setting should be lowered?

A Max tokens

B Temperature ✓ CORRECT

C Context window

D System message length

Why: Lowering temperature reduces randomness, making the model favor its single most likely, most consistent response.

DOMAIN 1 RECAP

The Big Picture So Far

1. **AI, ML, and Deep Learning:** Nested fields — ML learns from data, deep learning uses layered neural networks, generative AI creates new content.

2. **Responsible AI:** Six principles — Fairness, Reliability & Safety, Privacy & Security, Inclusiveness, Transparency, Accountability — "Fair Robots Protect Innocent Trusting Allies."

3. **Five AI workloads:** Text analysis, speech, computer vision, image generation, and information extraction.

4. **Models & agents:** Foundation models power generative AI; parameters like temperature shape output; agents extend chat into multi-step action.

That's Domain 1 (40–45% of the exam) fully covered. Next: putting it all into practice with Microsoft Foundry.

PART TWO

Implementing AI Solutions with Microsoft Foundry

The hands-on half of the exam: the Foundry portal, choosing and deploying models, prompt engineering, the SDK, and building agents.

Domain Weight: 55–60% of the exam

MEET THE PLATFORM

What Is Microsoft Foundry?

1. **Definition:** Microsoft's unified platform for discovering, building, evaluating, and deploying AI models, apps, and agents on Azure.

2. **Why it exists:** It brings model catalog browsing, prompt playgrounds, agent building, evaluation, and observability into one workspace.

3. **Model-agnostic by design:** Foundry hosts models from Microsoft, OpenAI, Meta, Mistral, and many other providers side by side.

4. **Exam framing:** Domain 2 is essentially “can you navigate and use Foundry correctly” — know the portal, not just the theory.

GETTING ORIENTED

Touring the Foundry Portal

1. **Project-based structure:** Work happens inside a Foundry project, which groups related models, deployments, and agents together.

2. **Model catalog:** A searchable library of foundation models you can browse, compare, and deploy.

3. **Playgrounds:** Interactive spaces to test prompts against a model before writing any code.

4. **Agents & Evaluation:** Dedicated areas to build agents and to measure model/agent quality with structured metrics.

PORTAL MAP

Key Foundry Navigation Areas

1 Models + Endpoints

Browse the catalog, deploy a model, and manage the live endpoints your app calls.

2 Playgrounds

Chat, prompt, or multimodal sandboxes to experiment before writing code.

3 Agents

Create, configure, and test AI agents with instructions, tools, and knowledge sources.

4 Evaluation & Observability

Score model/agent quality and monitor live usage, cost, and safety signals.

Browsing the Model Catalog

1. **What's in the catalog:** Hundreds of foundation models, filterable by provider, modality (text, vision, audio), size, and license.

2. **Model cards:** Each model lists its capabilities, context window, benchmark scores, and responsible-AI documentation.

3. **Try before you deploy:** You can test a candidate model directly in a playground before committing to a deployment.

4. **Exam cue:** The catalog is the starting point for nearly every Domain 2 scenario — "where would you first look to compare models?"

Picking the Right Model for the Job

1. **Task fit:** Match model modality and specialty to the task — a vision model for images, a language model for text.

2. **Capability vs. cost:** A larger, more capable model costs more per token and responds more slowly than a smaller one.

3. **Context window needs:** Long documents or long conversations require a model with a sufficiently large context window.

4. **Responsible AI fit:** Check the model card for known limitations, especially for sensitive or regulated use cases.

SIZE MATTERS

Small Language Models vs. Large Language Models

Small Language Model (SLM)

- Fewer parameters, smaller footprint
- Faster inference, lower cost per call
- Can run on-device or at the edge
- Best for narrow, well-defined tasks

Large Language Model (LLM)

- Billions of parameters, broad knowledge
- Stronger reasoning across varied tasks
- Typically cloud-hosted, higher cost
- Best for complex, open-ended tasks

GOING LIVE

Deploying a Model in Foundry

1. **Step 1:** Select a model from the catalog and review its card for capability and cost.

2. **Step 2:** Choose a deployment type — serverless pay-as-you-go, or managed compute with reserved capacity.

3. **Step 3:** Configure the deployment name, region, and any content-filter policy.

4. **Step 4:** Get the endpoint and key, then call it from the playground, SDK, or REST API.

PRACTICE CHECK 7

A team needs to quickly compare three different foundation models' output quality on the same sample prompt before choosing one. Where should they do this?

A. In the Foundry model catalog listing page only

B. In a Foundry playground ✓ CORRECT

C. By deploying all three to production first

D. In the Azure billing dashboard

Why: Playgrounds exist precisely so teams can test and compare prompts against different models interactively before committing to a deployment.

PROMPTING

Prompt Engineering Fundamentals

1. **Definition:** The practice of crafting input text to reliably guide a generative model toward the output you want.

2. **Why it matters:** The same model can produce dramatically different quality output depending purely on how the prompt is written.

3. **Core levers:** Clarity of instructions, relevant context, examples, and explicit output format.

4. **Exam framing:** AI-901 expects you to recognize good vs. weak prompts and to know the standard prompting techniques by name.

PROMPTING

System, User & Assistant Messages

1. **System message:** Sets the model's role, tone, and behavioral boundaries — defined once, applied for the whole conversation.

2. **User message:** The actual question or instruction from the person (or app) interacting with the model.

3. **Assistant message:** The model's own prior responses, kept in history so it can maintain a coherent conversation.

4. **Why the roles matter:** Foundry chat APIs are structured around this three-role message format — it's how conversation state is represented.

MEMORY BOOSTER

Recall the Chat Message Roles

“Silent Umpires Assist”

S	**System**	Sets the ground rules and persona before the conversation starts.
U	**User**	The person's (or app's) actual question or instruction.
A	**Assistant**	The model's own previous replies, kept for context.

“Silent Umpires Assist” — System sets rules, User asks, Assistant replies.

Core Prompting Techniques

1. **Zero-shot prompting:** Ask the model to perform a task with no examples — relying entirely on its pretrained knowledge.

2. **Few-shot prompting:** Provide a handful of example input/output pairs in the prompt to steer the style or format.

3. **Chain-of-thought prompting:** Ask the model to reason step-by-step before giving a final answer, improving accuracy on complex tasks.

4. **Prompt templates:** Reusable prompt structures with placeholders, so an app can generate consistent prompts at scale.

WITH OR WITHOUT EXAMPLES

Zero-Shot vs. Few-Shot Prompting

Zero-Shot

- No examples given in the prompt
- Fastest and cheapest to write
- Relies fully on the model's training
- Works well for common, general tasks

Few-Shot

- 2–5 examples included in the prompt
- Guides tone, format, and edge cases
- Uses more tokens per request
- Works well for niche or formatted output

Prompt Engineering Best Practices

1 Be specific

State the exact task, audience, tone, and desired length — vague prompts get vague answers.

2 Show, don't just tell

Include a short example of the exact output format you want.

3 Give the model an “out”

Instruct it to say “I don't know” rather than guess, reducing hallucination risk.

4 Iterate

Treat prompt writing as a testing loop — refine based on the model's actual responses.

GROUNDING

Grounding & Retrieval-Augmented Generation (RAG)

1. **The problem:** A model's built-in knowledge is frozen at training time and may not include your private or up-to-date data.

2. **Grounding:** Supplying a model with relevant, trustworthy source data at prompt time so its answer is based on facts, not guesses.

3. **RAG pattern:** Retrieve relevant documents from a knowledge source (like a search index), then inject them into the prompt before generation.

4. **Why it matters:** RAG is the standard way Foundry agents answer questions about your own documents accurately and reduce hallucination.

PRACTICE CHECK 8

A developer wants a support bot to answer questions using the company's internal 400-page product manual, which the base model has never seen. What's the best approach?

A. Increase the temperature parameter

B. Use retrieval-augmented generation to ground responses in the manual ✓ CORRECT

C. Switch to a smaller language model

D. Add a longer system message describing the manual's contents

Why: RAG retrieves the relevant passages from the actual manual at query time and grounds the model's answer in them — far more reliable than hoping the model "knows" private content.

Building a Chat App with the Foundry SDK

1. **What the SDK gives you:** Language-specific libraries (Python, C#, JavaScript, and more) to call Foundry-hosted models from your own code.

2. **Basic pattern:** Authenticate, create a client pointed at your project endpoint, send a chat message, and read back the completion.

3. **Why code instead of the playground:** The playground is for experimentation; the SDK is how you embed AI into a real, running application.

4. **Exam framing:** You won't need to memorize full syntax — you need to know the sequence of steps and key objects involved.

THE BASIC PATTERN

A Foundry Chat App Workflow

1

Initialize the Client

Authenticate and connect to your Foundry project or model endpoint.

2

Send the Prompt

Pass a structured message list (system, user) to the chat completion call.

3

Receive the Completion

The model returns generated text, plus usage and metadata.

4

Render the Result

Display the response in your app, and optionally store it in conversation history.

SDK CONCEPTS

Key Foundry SDK Objects

1. **Client:** The authenticated connection object your code uses to talk to a Foundry project or a specific model deployment.

2. **Messages array:** An ordered list of system/user/assistant turns representing the conversation sent with each request.

3. **Completion / Response:** The object returned by the model, containing the generated text and metadata like token usage.

4. **Deployment name:** The identifier of the specific deployed model your client call targets.

Function Calling & Tool Use

1. **Definition:** A capability where the model can request that your application run a specific function, then use the result in its answer.

2. **Why it's powerful:** It lets a model that can only generate text actually trigger real actions — look up live data, book a meeting, run a calculation.

3. **The pattern:** You describe available functions to the model; the model decides when to call one and with what arguments; your app executes it and returns the result.

4. **Foundation for agents:** Function calling / tool use is the core mechanism that turns a chat model into an agent.

TWO WAYS TO RECEIVE OUTPUT

Chat Completion vs. Streaming Responses

Standard Completion

- Waits for the full response before returning
- Simple to implement and parse
- Higher perceived latency for long answers
- Good for backend/batch processing

Streaming Response

- Returns tokens incrementally as generated
- Feels faster and more interactive to users
- Slightly more complex to implement
- Good for live chat interfaces

PRACTICE CHECK 9

A developer wants a chat interface where words appear on screen one at a time as the model “types,” rather than waiting for the full answer. What should they use?

A. A standard (non-streaming) chat completion call

B. A streaming response ✓ CORRECT

C. A higher max_tokens value

D. A fine-tuned model

Why: Streaming responses return generated tokens incrementally, which is exactly the “typing” effect described.

What Makes Up a Foundry Agent?

1. **An agent bundles four things:** A model, a set of instructions, a set of tools it can call, and optional knowledge sources to ground its answers.

2. **Why bundle them:** This packaging lets you define reusable, purpose-built assistants instead of re-writing prompt logic in every app.

3. **Persistent and stateful:** A Foundry agent can maintain conversation threads and reuse its configuration across many sessions.

4. **Exam framing:** Expect the exam to ask you to identify which of the four components is missing or misconfigured in a scenario.

THE FOUR BUILDING BLOCKS

Core Components of an Agent

1 Model

The underlying foundation model that powers the agent's reasoning and language generation.

2 Instructions

The system-level guidance defining the agent's role, tone, and constraints — like a job description.

3 Tools

Functions, APIs, or built-in capabilities (like code execution or file search) the agent can invoke.

4 Knowledge

Connected data sources — files, search indexes — the agent can retrieve from to ground its answers.

MEMORY BOOSTER

Recall the Agent Building Blocks

“My Ideas Take Knowledge”

M	**Model**	The foundation model that powers the agent's reasoning.
I	**Instructions**	The role, tone, and constraints the agent must follow.
T	**Tools**	Functions and APIs the agent is allowed to call.
K	**Knowledge**	Connected data the agent retrieves to ground its answers.

“My Ideas Take Knowledge” — Model, Instructions, Tools, Knowledge: the four pillars of every Foundry agent.

Creating & Testing an Agent

1. **Step 1 — Define:** Pick a model and write clear instructions describing the agent's purpose and boundaries.

2. **Step 2 — Equip:** Attach the tools and knowledge sources the agent needs to complete its job.

3. **Step 3 — Test:** Use the agent playground to run sample conversations and confirm it behaves as expected.

4. **Step 4 — Iterate:** Refine instructions or tools based on test results before connecting the agent to a real application.

AGENT TOOLS

Common Built-In Agent Tools

1. **Code Interpreter:** Lets the agent write and execute code to perform calculations, analyze data, or generate charts.

2. **File Search:** Lets the agent search and retrieve relevant passages from uploaded documents — a built-in RAG capability.

3. **Custom Functions:** Developer-defined functions the agent can call, connecting it to your own APIs or business logic.

4. **Bing / Web Grounding:** Lets the agent retrieve current information from the web to answer time-sensitive questions.

SCALING UP

Multi-Agent Orchestration & Workflows

1. **The idea:** Complex tasks are broken into steps, each handled by a specialized agent, coordinated toward one overall goal.

2. **Why not one giant agent:** Specialized agents with narrow instructions are more reliable and easier to test and maintain than one that does everything.

3. **Orchestration pattern:** A coordinating agent (or workflow) routes tasks to the right specialist agent and combines their results.

4. **Exam cue:** A scenario describing several distinct sub-tasks handled by different specialists points to multi-agent orchestration.

SEEING IT IN ACTION

Example: A Multi-Agent Workflow

1

Intake Agent

Receives the customer request and classifies its intent.

2

Research Agent

Uses file search / web grounding to gather relevant facts.

3

Drafting Agent

Writes a response using the gathered research.

4

Review Agent

Checks the draft against policy before it's sent to the customer.

Each agent has one narrow job — this is what makes multi-agent workflows reliable and easy to debug.

PRACTICE CHECK 10

You're building a Foundry agent that must answer questions strictly from a company's uploaded policy PDFs and refuse anything outside that scope. Which two components are most critical to configure carefully?

A Model size and max tokens only

B Instructions and Knowledge (file search) ✓ CORRECT

C Temperature and streaming mode

D Deployment region and billing tier

Why: Instructions define the scope/refusal behavior, while Knowledge (via file search) grounds every answer in the uploaded policy documents — together they enforce the required scope.

GOVERNANCE IN PRACTICE

Content Safety & Content Filters

1. **What content filters do:** Automatically screen prompts and completions for categories like violence, hate, sexual content, and self-harm.

2. **Configurable severity levels:** Foundry lets you tune how strict each filter category is, based on your application's audience and use case.

3. **Applied at two points:** Filters can screen the incoming user prompt and the outgoing model completion.

4. **Exam cue:** Content filters are the direct, practical implementation of the Reliability & Safety principle inside Foundry.

MEASURING QUALITY

Evaluation in Foundry

1. **Why evaluate:** You need objective evidence a model or agent performs well before — and after — it goes live.

2. **Built-in metrics:** Groundedness, relevance, coherence, fluency, and safety scores, computed automatically against test data.

3. **Custom evaluators:** You can define your own criteria specific to your business scenario.

4. **Red-teaming:** Deliberately probing a model or agent with adversarial prompts to surface safety weaknesses before release.

WATCHING IT RUN

Observability & Monitoring

1. **Why it matters:** A model that performed well in testing can still drift or misbehave once real users start interacting with it.

2. **What you monitor:** Request volume, latency, token usage/cost, error rates, and safety-filter trigger rates.

3. **Tracing:** Following a single request through every step — retrieval, tool calls, generation — to debug agent behavior.

4. **Exam cue:** “We need to know why our agent gave a wrong answer last Tuesday” is an observability/tracing scenario.

PROTECTING THE SYSTEM

Security: Identity & Access vs. Data Protection

Identity & Access

- Role-based access control (RBAC) on projects
- Managed identities instead of hardcoded keys
- API keys rotated and scoped narrowly
- Network restrictions on who can call an endpoint

Data Protection

- Encryption of data at rest and in transit
- Data not used to retrain shared/public models
- Sensitive data masked before logging or tracing
- Compliance with regional data residency rules

RUNNING IT EFFICIENTLY

Cost & Resource Management

1 Token-based billing

Serverless models are billed by input and output tokens consumed — shorter prompts and responses cost less.

2 Right-sizing the model

Using an oversized model for a simple task wastes cost and latency — match model size to task complexity.

3 Quota & rate limits

Foundry deployments have configurable limits to control throughput and protect budgets.

4 Caching & reuse

Reusing prior results for repeated queries reduces redundant model calls and cost.

Supportive Azure AI Services

1. **Azure AI Search:** A search and indexing service commonly used as the retrieval half of a RAG pipeline.

2. **Azure Bot Service:** A framework for building, connecting, and deploying conversational bots across channels like Teams and web chat.

3. **Azure AI Content Understanding:** Extracts structured insight from multimodal content — documents, images, audio, and video — inside Foundry.

4. **Exam cue:** These services complement Foundry rather than replace it — know what each one is for at a high level.

PRACTICE CHECK 11

After deploying an agent, a team wants to understand exactly why it produced a specific wrong answer during a live customer chat last week. What should they check first?

A The model catalog listing

B Observability / tracing logs for that request ✓ CORRECT

C The content filter severity settings only

D The billing and cost dashboard

Why: Observability and tracing let you replay a specific request's full path — retrieval, tool calls, and generation — to diagnose exactly what went wrong.

DOMAIN 2 RECAP

The Big Picture: Microsoft Foundry

1. **Foundry is the hub:** Model catalog, playgrounds, agents, evaluation, and observability all live in one workspace.

2. **Prompting is a skill:** System/user/assistant roles, zero-shot vs. few-shot, and grounding via RAG all shape output quality.

3. **The SDK operationalizes it:** Client → send prompt → receive completion → render — with function calling as the bridge to real actions.

4. **Agents = Model + Instructions + Tools + Knowledge:** And governance — content filters, evaluation, and observability — keeps everything safe and accountable.

That's Domain 2 (55–60% of the exam) fully covered. Next: locking it all in for exam day.

MEMORY BOOSTER

The Master Map of AI-901

“Concepts First, Then Foundry”

C	**Concepts (Domain 1)**	AI/ML foundations, Responsible AI, and the five core AI workloads.
F	**Foundry (Domain 2)**	The portal, model deployment, prompting, the SDK, and building agents.

If you can explain a concept AND show where it lives in Foundry, you're exam-ready for that topic.

FLASHCARD WALL 1

Vocabulary Rapid Review — Concepts

1 Hallucination

Confident but factually incorrect model output — mitigated by grounding and lower temperature.

2 Bias

Systematic unfair skew in model output, usually traced back to unrepresentative training data.

3 Context Window

The maximum tokens of text a model can consider at once, prompt plus history included.

4 Multimodal

A model that can process more than one type of input/output — e.g., text and images together.

5 Inference

Running a trained model on new input to produce a prediction or generated output.

6 Prompt

The input text given to a generative model to elicit a response.

FLASHCARD WALL 2

Vocabulary Rapid Review — Foundry

1 Playground

An interactive Foundry space to test prompts against a model before writing code.

2 Endpoint

The network address your application calls to send requests to a deployed model.

3 Grounding

Supplying real, trustworthy data to a model at prompt time to improve factual accuracy.

4 Tool / Function Calling

Letting a model trigger real application code or APIs as part of its response.

5 Evaluation Metric

A scored measure — like groundedness or relevance — used to judge model or agent quality.

6 Red-Teaming

Deliberately probing a system with adversarial prompts to find safety weaknesses.

PRACTICE CHECK 12

Which statement correctly distinguishes a foundation model from a fine-tuned model?

A A foundation model is always smaller than a fine-tuned model

B A fine-tuned model is further trained on a narrower dataset to specialize a foundation model ✓ CORRECT

C Fine-tuning changes only the system message, not the model itself

D Foundation models cannot be used without fine-tuning

Why: Fine-tuning takes a general-purpose foundation model and continues training it on a smaller, specific dataset to specialize its behavior, tone, or format.

PRACTICE CHECK 13

A healthcare scheduling agent must never book an appointment without a human confirming it first. Which Responsible AI principle does this safeguard primarily support?

A Inclusiveness

B Transparency

C Accountability ✓ CORRECT

D Privacy & Security

Why: Requiring human confirmation before an irreversible action keeps a person responsible for the outcome — the core of Accountability.

AVOID THESE TRAPS

Common Exam Traps & Tricky Distinctions

1. **Classification vs. Detection:** Classification labels the whole image; detection locates and boxes multiple objects within it.

2. **Fairness vs. Inclusiveness:** Fairness is about equal treatment/outcomes; Inclusiveness is about who can access and use the system at all.

3. **Grounding vs. Fine-tuning:** Grounding supplies fresh data at prompt time; fine-tuning permanently updates the model's weights.

4. **Zero-shot vs. Few-shot:** Zero-shot gives no examples; few-shot includes sample input/output pairs in the prompt.

5. **Temperature vs. Max Tokens:** Temperature controls creativity/randomness; max tokens controls response length, not quality.

KNOW THE HISTORY

AI-900 vs. AI-901: What Changed

AI-900 (Retired June 2026)

- Focused on describing AI concepts
- Centered on Azure Cognitive Services
- Little to no coding expected
- Lighter on generative AI and agents

AI-901 (Current Exam)

- Concepts plus hands-on implementation
- Centered on Microsoft Foundry
- Basic Python/SDK familiarity expected
- Strong focus on generative & agentic AI

GAME DAY

Your Exam Day Plan

1

Sleep & Review

Skim mnemonic slides the night before — don't cram brand-new material.

2

Arrive Early

Check in with ID at least 15 minutes ahead of your Pearson VUE appointment.

3

Triage Questions

Answer confident questions first; flag and return to uncertain ones.

4

Reason, Don't Panic

Eliminate clearly wrong options first — most items narrow to two real choices.

FINAL TIPS

Last-Mile Study Advice

1. **Practice inside Foundry itself:** Free-tier hands-on time in the actual portal cements concepts far better than reading alone.

2. **Say the mnemonics aloud:** Verbalizing — not just reading — dramatically improves recall under exam pressure.

3. **Re-take the practice checks cold:** Do them again 24 hours later without looking at the explanations first.

4. **Don't neglect Responsible AI:** It's woven through both domains, not confined to one section — expect it in Foundry scenarios too.

ONE LAST LOOK

The Numbers That Matter

45–60%

Foundry Domain

The larger, hands-on half of the exam

40–45%

Concepts Domain

AI/ML fundamentals and Responsible AI

6

RAI Principles

“Fair Robots Protect Innocent Trusting Allies”

700/1000

Passing Score

Your target scaled score on exam day

MEMORY BOOSTER

The Complete Memory Wall

Every Mnemonic From This Guide, In One Place

1	**ML Types**	“Sam Uses Rewards” — Supervised, Unsupervised, Reinforcement.
2	**Responsible AI**	“Fair Robots Protect Innocent Trusting Allies” — F-R-P-I-T-A.
3	**Text Analysis**	“Keep Every Sentence Loud & True” — Key phrase, Entity, Sentiment, Language, Translate/Summarize.
4	**Computer Vision**	“Clever Owls Detect Objects” — Classification, OCR, Detection, (Facial) Overview.
5	**Model Parameters**	“Tame The Model's Speech” — Temperature, Top P, Max tokens, System message.
6	**Chat Roles**	“Silent Umpires Assist” — System, User, Assistant.
7	**Agent Components**	“My Ideas Take Knowledge” — Model, Instructions, Tools, Knowledge.

Seven phrases. Every major list on the exam. Review this one slide the morning of your test.

YOU'RE READY

You're Ready for AI-901.

Concepts learned. Foundry practiced. Mnemonics locked in.

- Revisit the gold mnemonic slides one more time before test day.
- Get hands-on in the real Microsoft Foundry portal if you haven't already.
- Trust your preparation — reason through each question methodically.
- After passing: explore Azure AI Engineer Associate as your next step.

YOU'RE READY

Good Luck — Go Earn That Certification.

Microsoft Certified: Azure AI Fundamentals

- This guide covered both exam domains in full: Concepts (40–45%) and Foundry (55–60%).
- Keep this deck as your final-review companion in the days before your exam.
- Official exam updates: learn.microsoft.com/certifications/azure-ai-fundamentals
- Thank you for studying with this guide — now go pass that exam.